

Data Science Case Challenge: LoanTap Credit Risk Prediction

Business Context

LoanTap is a new-age digital lending platform offering customized credit products like personal loans, salary advances, and overdraft facilities to salaried professionals and MSMEs. With growing customer volumes, the company aims to build a **data-driven credit underwriting model** that automates risk evaluation while keeping Non-Performing Assets (NPAs) low.

LoanTap's leadership has tasked its Data Science team to identify which applicants are most likely to default, and which can be safely extended credit. This model will form the foundation for future risk-based pricing and customer segmentation.

Problem Statement (For Learners)

As a **Data Scientist at LoanTap**, your objective is to develop a **Logistic Regression model** that predicts whether a borrower will **fully repay** the loan or **default**.

The project requires you to perform: 1. **Exploratory Data Analysis (EDA)** – Understand the distribution, trends, and relationships among financial, demographic, and behavioral variables. 2. **Feature Engineering** – Create new indicators such as flags for bankruptcies or mortgage accounts that may influence credit risk. 3. **Data Preprocessing** – Handle missing values, outliers, scaling, and categorical encoding. 4. **Model Building** – Train a Logistic Regression model and interpret the coefficients. 5. **Evaluation** – Assess model performance using ROC-AUC, Precision, Recall, and F1-Score. 6. **Business Interpretation** – Recommend strategies to minimize NPAs while maintaining lending opportunities.

Dataset Description

File: https://drive.google.com/file/d/1ZPYj7CZCfxntE8p2Lze_4Q04MyEOy6_d/view

Each record represents one applicant, containing demographic, employment, credit history, and loan-related variables.

Column	Description
loan_amnt	The amount of loan applied for by the borrower
term	Loan tenure in months (36 or 60)
int_rate	Annual interest rate (%)
installment	Monthly installment amount
grade	LoanTap's assigned risk grade

Column	Description
emp_length	Employment length in years (0–10)
home_ownership	Home ownership status (OWN, RENT, MORTGAGE)
annual_inc	Annual income declared by borrower
verification_status	Income verification status
purpose	Purpose of the loan (e.g., debt consolidation, home improvement)
dti	Debt-to-income ratio
revol_bal	Revolving credit balance
pub_rec	Number of public derogatory records
mort_acc	Number of mortgage accounts
pub_rec_bankruptcies	Number of bankruptcy records
loan_status	Target variable (Fully Paid / Charged Off)

Evaluation Metrics

Learners will be evaluated on: 1. **EDA & Data Cleaning (20%)** – Handling missing data, detecting outliers, variable understanding. 2. **Feature Engineering (15%)** – Creating meaningful derived variables. 3. **Modeling (20%)** – Building and tuning the Logistic Regression model. 4. **Model Evaluation (25%)** – Reporting ROC-AUC, Precision, Recall, and F1 Score. 5. **Business Interpretation (20%)** – Explaining results and making actionable credit risk recommendations.

Discussion Prompts for Learners

- What percentage of applicants have fully repaid their loans?
- How does interest rate influence loan defaults?
- Which factors (income, grade, dti) have the strongest predictive power?
- How should the business choose a decision threshold between Precision and Recall?
- What would be the financial implication of increasing recall at the cost of precision?

Deliverables

- A Jupyter Notebook (LoanTap_Logistic_Regression.ipynb) with code, visuals, and commentary.
- A concise report or presentation summarizing findings, metrics, and recommendations.

